

TCP Segment structure=Source Port Address: A 16-bit field that holds the port address of the application that is sending the data segment. **Destination Port Address:** A 16-bit field that holds the port address of the application in the host that is receiving the data segment. **Sequence Number:** A 32-bit field that holds the sequence number, i.e, the byte number of the first byte that is sent in that particular segment. It is used to reassemble the message at the receiving end of the segments that are received out of order. **Acknowledgement Number:** A 32-bit field that holds the acknowledgement number, i.e, the byte number that the receiver expects to receive next. It is an acknowledgement for the previous bytes being received successfully. **Header Length (HLEN):** This is a 4-bit field that indicates the length of the TCP header by a number of 4-byte words in the header, i.e if the header is 20 bytes(min length of TCP header), then this field will hold 5 (because $5 \times 4 = 20$) and the maximum length: 60 bytes, then it'll hold the value 15(because $15 \times 4 = 60$). Hence, the value of this field is always between 5 and 15. **Control flags:** These are 6 1-bit control bits that control connection establishment, connection termination, connection abortion, flow control, mode of transfer etc.

[Internet Protocol Version 4 \(IPv4\)](#) is the fourth version of the Internet Protocol.It is a connectionless protocol used in packet-switched networks.IPv4 is widely used for data communication over different types of networks, such as [Ethernet](#).It provides logical addressing to uniquely identify each device on a network.**IPv4 Datagram Header=VERSION:** Version of the IP protocol (4 bits), which is 4 for IPv4 **HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15. **Type of service:** Low Delay, High Throughput, Reliability (8 bits) **Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes. **Identification:** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits) **Flags:** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order) **Fragment Offset:** Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. **Source IP address:** 32 bits IP address of the sender **Destination IP address:** 32 bits [IP address](#) of the receiver

The next generation Internet Protocol (IP) address standard, known as IPv6, is meant to work in cooperation with [IPv4](#). To communicate with other devices, a computer, smartphone, home automation component, Internet of Things sensor, or any other Internet-connected device needs a numerical IP address.**types Unicast Addresses :** Only one interface is specified by the unicast address. A packet moves from one host to the destination host when it is sent to a unicast address destination.**Multicast Addresses:** It represents a group of IP devices and can only be used as the destination of a datagram.**Anycast Addresses:** The multicast address and the anycast address are the same. The way the anycast address varies from other addresses is that it can deliver the same IP address to several servers or devices.

Services tcp=Process-to-Process Communication: TCP enables process-to-process communication using 16-bit port numbers to identify the sending and receiving processes.**2. Stream oriented:** TCP sends data as a byte stream, grouping it into segments with headers, which are then encapsulated into IP packets.**3. Full-duplex service:** This means that the communication can take place in both directions at the same time. **4.Connection-oriented service:** Unlike UDP, TCP provides a connection-oriented service. It defines 3 different phases: Connection establishment.Data transfer.Connection termination**5. Reliability:** TCP ensures reliability through checksums, retransmissions, acknowledgements, sequencing, and congestion control.**6. Multiplexing:** TCP does multiplexing and de-multiplexing at the sender and receiver ends respectively as a number of logical connections can be established between port numbers over a physical connection.

[Carrier Sense Multiple Access](#) with Collision Detection (CSMA/CD) is a [network protocol](#) for carrier transmission that operates in the Medium Access Control ([MAC](#)) layer. It senses or listens whether the shared channel for transmission is busy or not, and defers transmissions until the channel is free. The [collision detection](#) technology detects collisions by sensing transmissions from other stations. On detection of a collision, the station stops transmitting, sends a jam signal, and then waits for a random time interval before retransmission.**types1-Persistent:** It senses the shared channel first and delivers the data right away if the channel is idle. It is an aggressive transmission algorithm.**Non-Persistent:** It first assesses the channel before transmitting data; if the channel is idle, the node transmits data right away. **P-Persistent:** It consists of the 1-Persistent and Non-Persistent modes combined. Each node observes the channel in the 1-Persistent mode and if the channel is idle, it sends a frame with a P probability.

Flow control in networking manages the rate of data transmission between devices to ensure efficient processing and prevent data overflow. When a device receives more data than it can handle, it risks losing information or needing retransmission.**Stop and Wait ARQ**The sender sends the packet and waits for the acknowledgment of the packet. Once the acknowledgment reaches the sender, it transmits the next packet in a row. If the acknowledgment does not reach the sender before the specified time, known as the timeout, the sender sends the same packet again. **Go Back N ARQ**The sender sends N packets which are equal to the window size. Once the entire window is sent, the sender then waits for a cumulative acknowledgement to send more packets. On the receiver end, it receives only in-order packets and discards out-of-order packets. As in case of packet loss, the entire window would be re-transmitted. **Selective Repeat ARQ**The sender sends packets of window size N and the receiver acknowledges all packets whether they were received in order or not. In this case, the receiver maintains a buffer to contain out-of-order packets and sorts them. The sender selectively re-transmits the lost packet and moves the window forward.